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GATE ECE · SIGNALS AND SYSTEMS · CHAPTER-WISE PYQS

Discrete Time Signal Fourier Series Fourier Transform

GATE ECE Signals and Systems · Previous Year Questions with Answer Key

14	10	2005 – 2024	English
QUESTIONS	PAPERS COVERED	YEARS	LANGUAGE

INSIDE THIS PDF

Marks 1	3 questions
Marks 2	11 questions
Answer key	at the end

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Marks 1 3 questions

Q1. GATE ECE 2015 Set 3

The value of $\sum_{n=0}^{\infty} n\left(\frac{1}{2}\right)^n$ is _____.

Open solution

Q2. GATE ECE 2014 Set 4

A Fourier transform pair is given by $\left(\frac{2}{3}\right)^n u[n+3] \overset{FT}{\Leftrightarrow} \frac{Ae^{-j6\pi f}}{1-\left(\frac{2}{3}\right)e^{-j2\pi f}}$,

where u(n) denotes the unit step sequence. The value of A is_____.

Open solution

Q3. GATE ECE 2005

Let x(n) = $\left(\frac{1}{2}\right)^n$ u(n), y(n) = x^2, and Y ($(e^{j\omega})$ be the Fourier transform of y(n). Then Y ((e^{jo}) is

(A) $\frac{1}{4}$ (B) 2 (C) 4 (D) $\frac{4}{3}$

Open solution

Marks 2 11 questions

Q4. GATE ECE 2024

The radian frequency value(s) for which the discrete time sinusoidal signal $x[n] = A\cos(\Omega n + \pi/3)$ has a period of 40 is/are __.

(A) 0.15π (B) 0.225π (C) 0.3π (D) 0.45π

Open solution

Q5. GATE ECE 2023

Consider a discrete-time periodic signal with period N = 5. Let the discrete-time Fourier series (DTFS) representation be $x[n] = \sum_{k=0}^{4} a_k e^{\frac{jk2\pi m}{5}}$, where $a_0 = 1, a_1 = 3j, a_2 = 2j, a_3 = -2j$ and $a_4 = -3j$.

The value of the sum $\sum_{n=0}^{4} x[n]\sin\frac{4\pi n}{5}$ is

(A) -10 (B) 10 (C) -2 (D) 2

Open solution

Q6. GATE ECE 2021

Consider the signals $x[n] = 2^{n-1}u[-n+2]$ and $y[n] = 2^{-n+2}u[n+1]$, where $u[n]$ is the unit step sequence. Let $X\left(e^{j\omega}\right)$ and $Y\left(e^{j\omega}\right)$ be the discrete-time Fourier transform of $x[n]$ and $y[n]$, respectively. The value of the integral

$\frac{1}{2\pi}\int_o^{2\pi} X\left(e^{j\omega}\right)Y\left(e^{-j\omega}\right)d\omega$

(round off to one decimal place) is ____

Open solution

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Q7. GATE ECE 2018

Let X[k] = k + 1, 0 ≤ k ≤ 7 be 8-point DFT of a sequence x[n],

where X[k] = $\sum_{n=0}^{N-1} x[n]e^{-j2\pi nk/N}$.

The value (correct to two decimal places) of $\sum_{n=0}^{3} x[2n]$ is _____.

Open solution

Q8. GATE ECE 2017 Set 1

Let h[n] be the impulse response of a discrete time linear time invariant (LTI) filter. The impulse response is given by h[0] = $\frac{1}{3}; h[1] = \frac{1}{3}; h[2] = \frac{1}{3}; and\, h[n]$ =0 for n < 0 and n > 2. Let H (ω) be the Discrete-time Fourier transform (DTFT) of h[n], where ω is the normalized angular frequency in radians. Given that H(ω_o) = 0 and 0 < ω_0 < π, the value of ω_o (in radians) is equal to _____.

Open solution

Q9. GATE ECE 2017 Set 1

Two discrete-time signals x [n] and h [n] are both non-zero only for n = 0, 1, 2 and are zero otherwise. It is given that x(0)=1, x[1] = 2, x [2] =1, h[0] = 1, let y [n] be the linear convolution of x[n] and h [n]. Given that y[1]= 3 and y [2] = 4, the value of the expression (10y[3] +y[4]) is _____.

Open solution

Q10. GATE ECE 2016 Set 1

Consider the signal $x[n] = 6\delta[n+2] + 3\delta[n+1] + 8\delta[n] + 7\delta[n-1] + 4\delta[n-2]$.
If X($e^{t\omega}$)is the discrete-time Fourier transform of x[n],

then $\frac{1}{\pi}\int_{-\pi}^{\pi} X(e^{j\omega})\sin^2(2\omega)d\omega$ is equal to _____.

Open solution

Q11. GATE ECE 2015 Set 3

Let $\widetilde{x}[n] = 1 + \cos\left[\frac{\pi n}{8}\right]$ be a periodic signal with period 16. Its DFS coefficients are defined by $a_k = \frac{1}{16}\sum_{n=0}^{15} \widetilde{x}[n]\exp\left(-j\frac{\pi}{8}kn\right)$ for all k. The value of the coeffcients a_{31} is _____.

Open solution

Q12. GATE ECE 2007

A 5-point sequence x [n] is given as x[−3] =1, x[−2] =1, x[−1] =0, x[−0] = 5, x[−1] = 1. Let X($e^{j\omega}$) denote the discrete - time Fourier transform of x(n). The value of $\int_{-\pi}^{\pi} x((e^{j\omega})\, d\omega$ is

(A) 5 (B) 10π (C) 16π (D) 5+ j 10 π

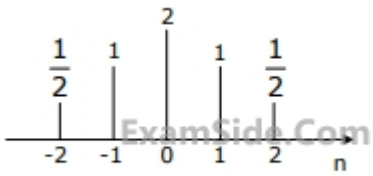
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Q13. GATE ECE 2005

Open solution

A sequence x(n) has non-zero values as shown in Fig.

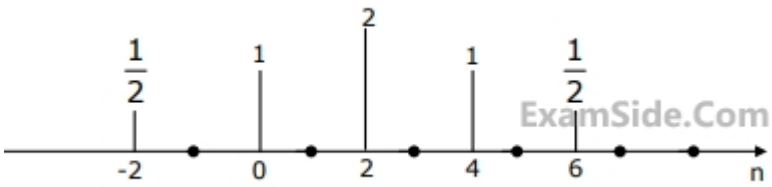


The sequence

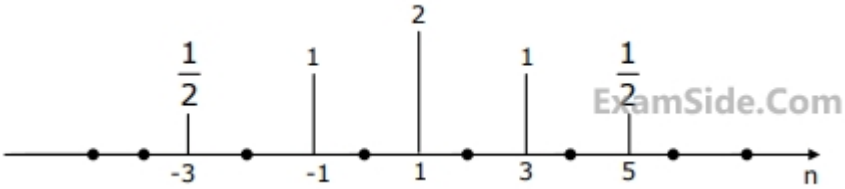
$$y(n) = \begin{cases} x\left(\frac{n}{2} - 1\right) & for\ n\ even \\ 0 & for\ n\ odd \end{cases}$$

will be

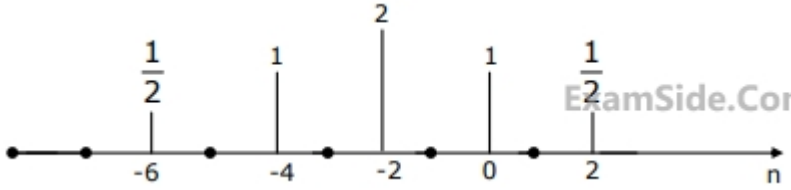
(A)



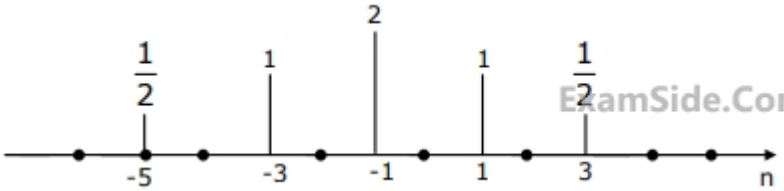
(B)



(C)



(D)

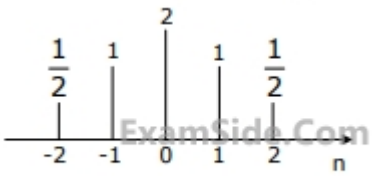


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Q14. GATE ECE 2005

Open solution

A sequence x(n) has non-zero values as shown in figure. 1



The Fourier transform of y(2n) will be

(A) $e^{-j2\omega}\left[\cos 4\omega + 2\cos 2\omega + 2\right]$

(B) $\left[\cos 2\omega + 2\cos\omega + 2\right]$

(C) $e^{-j\omega}\left[\cos 2\omega + 2\cos\omega + 2\right]$

(D) $e^{j2\omega}\left[\cos 2\omega + 2\cos\omega + 2\right]$

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Answer Key

Marks 1

1. **2**	2. **3.375**	3. **D**

Marks 2

4. **A, C, D**	5. **A**	6. **8**	7. **2.90 to 3.10**	8. **2.09**
9. **31**	10. **8**	11. **0.5**	12. **B**	13. **A**
14. **C**				

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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